WORKED SOLUTION

STEP III 2020, Question 5

Pure • Partial Fractions

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Read the question before you go on. It is in the STEP III paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

The opening identity is the finite geometric series in disguise, and it is exactly what turns $(1-(x/k)^n)/(x-k)$ into an explicit polynomial. That gives a complete partial-fraction decomposition of $1/(x^n(x-k))$ without any heavy comparing of coefficients. Differentiating then produces the decomposition with a *repeated* factor $(x-k)^2$, and the final integral is just a special case in which every term can be integrated directly.

Solution

The opening identity

Expand the right-hand side and split it into two sums:

$$\begin{aligned}(x-y)\sum_{r=1}^{n} x^{n-r}y^{r-1} &= \sum_{r=1}^{n} x^{n-r+1}y^{r-1} - \sum_{r=1}^{n} x^{n-r}y^{r} \\ &= (x^n + x^{n-1}y + \cdots + xy^{n-1}) - (x^{n-1}y + \cdots + xy^{n-1} + y^n) \\ &= x^n - y^n,\end{aligned}$$

since every term except x^n (from $r=1$ in the first sum) and y^n (from $r=n$ in the second) appears once in each bracket and cancels.

Part (i)(a)

Multiplying $\mathrm{F}(x) = \dfrac{A}{x-k} + \dfrac{\mathrm{f}(x)}{x^n}$ through by $x^n(x-k)$ gives

$$1 = Ax^n + \mathrm{f}(x)(x-k).$$

This is an identity between polynomials, true for all $x \neq 0, k$ and hence for all x (two polynomials that agree at infinitely many points are equal). Putting $x = k$ gives $1 = Ak^n$,

so $A = 1/k^n$ (allowed, since $k \neq 0$). Then

$$\mathrm{f}(x)(x-k) = 1 - \frac{x^n}{k^n}, \qquad \text{so} \qquad \mathrm{f}(x) = \frac{1}{x-k}\left(1 - \left(\frac{x}{k}\right)^n\right)$$

for $x \neq k$, as required.

Ask yourself How do we turn this into something without $x-k$ in the denominator?

Use the opening identity with 1 in place of x and x/k in place of y:

$$1 - \left(\frac{x}{k}\right)^n = \left(1 - \frac{x}{k}\right)\sum_{r=1}^{n}\left(\frac{x}{k}\right)^{r-1} = -\frac{x-k}{k}\sum_{r=1}^{n}\frac{x^{r-1}}{k^{r-1}}.$$

Dividing by $x-k$,

$$\mathrm{f}(x) = -\frac{1}{k}\sum_{r=1}^{n}\frac{x^{r-1}}{k^{r-1}}, \qquad \text{so} \qquad \frac{\mathrm{f}(x)}{x^n} = -\frac{1}{k}\sum_{r=1}^{n}\frac{1}{k^{r-1}x^{n+1-r}}.$$

(This also confirms that f really is a polynomial.) Now re-index with $s = n+1-r$, which runs over $1,\ldots,n$ as r does; then $r-1 = n-s$, and the sum becomes $\sum_{s=1}^{n} 1/(k^{n-s}x^s)$. Therefore

$$\mathrm{F}(x) = \frac{1}{k^n(x-k)} - \frac{1}{k}\sum_{r=1}^{n}\frac{1}{k^{n-r}x^r}.$$

Part (i)(b)

Since $x^n\mathrm{F}(x) = \dfrac{1}{x-k}$, differentiating both sides with the product rule gives

$$nx^{n-1}\mathrm{F}(x) + x^n\mathrm{F}'(x) = -\frac{1}{(x-k)^2}.$$

Dividing by $-x^n$ (fine, as $x \neq 0$):

$$\frac{1}{x^n(x-k)^2} = -\frac{n}{x}\mathrm{F}(x) - \mathrm{F}'(x).$$

From part (a),

$$-\frac{n}{x}\mathrm{F}(x) = -\frac{n}{xk^n(x-k)} + \sum_{r=1}^{n}\frac{n}{k^{n+1-r}x^{r+1}},$$

and differentiating the result of part (a) term by term,

$$-\mathrm{F}'(x) = \frac{1}{k^n(x-k)^2} - \frac{1}{k}\sum_{r=1}^{n}\frac{r}{k^{n-r}x^{r+1}} = \frac{1}{k^n(x-k)^2} - \sum_{r=1}^{n}\frac{r}{k^{n+1-r}x^{r+1}}.$$

Adding, the two sums combine to give

$$\frac{1}{x^n(x-k)^2} = \frac{1}{k^n(x-k)^2} - \frac{n}{xk^n(x-k)} + \sum_{r=1}^{n}\frac{n-r}{k^{n+1-r}x^{r+1}},$$

which is the required result.

Part (ii)

Take $n = 3$ and $k = 1$ in part (b). The $r = 3$ term of the sum vanishes, and

$$\frac{1}{x^3(x-1)^2} = \frac{1}{(x-1)^2} - \frac{3}{x(x-1)} + \frac{2}{x^2} + \frac{1}{x^3}.$$

The middle term needs one more split: $\dfrac{1}{x(x-1)} = \dfrac{1}{x-1} - \dfrac{1}{x}$. On $[2, N]$ everything is continuous and $x - 1 > 0$, so an antiderivative there is

$$G(x) = -\frac{1}{x-1} - 3\ln\left(\frac{x-1}{x}\right) - \frac{2}{x} - \frac{1}{2x^2},$$

and

$$\int_2^N \frac{1}{x^3(x-1)^2}\,\mathrm{d}x = G(N) - G(2).$$

? **Ask yourself** Why must we keep the two logarithms together?

Separately, $-3\ln(N-1)$ and $+3\ln N$ each tend to infinity, so we can't take the limit term by term. Combined, $\ln((N-1)/N) = \ln(1 - \frac{1}{N}) \to \ln 1 = 0$ as $N \to \infty$, because ln is continuous at 1. The other three terms of $G(N)$ clearly tend to 0, so $G(N) \to 0$.

Finally,

$$G(2) = -1 - 3\ln\tfrac{1}{2} - 1 - \tfrac{1}{8} = 3\ln 2 - \tfrac{17}{8},$$

so

$$\lim_{N\to\infty} \int_2^N \frac{1}{x^3(x-1)^2}\,\mathrm{d}x = \frac{17}{8} - 3\ln 2.$$

Checks

- The answer must be positive, since the integrand is positive: indeed $\frac{17}{8} - 3\ln 2 \approx 2.1250 - 2.0794 = 0.0456$.
- It should also be small: the integrand is $\frac{1}{8}$ at $x = 2$ and falls off roughly like x^{-5}. Crudely bounding it by $\frac{1}{8}(2/x)^5$ gives at most $\int_2^\infty 4x^{-5}\,\mathrm{d}x = \frac{1}{16}$. A numerical integration confirms the value $0.045558\ldots$ to many decimal places.
- Try $n = 1$ in part (b). The sum vanishes (its only term has $n - r = 0$), and the right-hand side is $\dfrac{1}{k(x-k)^2} - \dfrac{1}{xk(x-k)} = \dfrac{x-(x-k)}{xk(x-k)^2} = \dfrac{1}{x(x-k)^2}$, as it should be.

Common slips

- Getting the sign wrong when writing $1 - x/k$ as $-(x-k)/k$.

- Forgetting to re-index the sum in part (a), and ending up with powers of k and x that do not match the target.
- Splitting $\ln\big((x-1)/x\big)$ into two logarithms and then claiming that the integral diverges, or writing "$\infty - \infty = 0$".

Beyond STEP

What you have built is the general partial-fraction decomposition for factors of the form x^n and $(x-k)^m$; differentiating with respect to x (or with respect to the root k) is a standard way to generate the repeated-factor terms. Deciding whether an integral over an infinite range converges, as in part (ii), is the subject of improper integrals in a first analysis course. Abbott's *Understanding Analysis* treats these limits carefully.

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